

1 new descent/ascent

Let S be a numerical semigroup with the usual f, m, g, σ, r and Apéry set $\{w_0, \dots, w_{m-1}\}$ indexed by residue. The four operators below each return a new numerical semigroup.

1.1 DescentFlip

Walk strata $l = 0, 1, \dots, \text{level} - 1$. At the first stratum that contains an Apéry element w of S with $w \neq f + m$, pick the largest such w and add $w - m$ as a generator (this is a gap of S , in the same residue class as w). No-op if $f < m$ or no such w exists.

```
pub fn descent_flip(&self) -> Self {
    if self.f < self.m {
        return self.clone();
    }
    let mut newgen = self.gen_set.clone();
    let amu = self.f + self.m;
    for l in 0..self.level {
        if let Some(largest) = self
            .apery_set
            .iter()
            .filter(|&x| x + l * self.m > self.f && x + l * self.m < amu)
            .max()
        {
            newgen.push(*largest - self.m);
            return compute(&newgen);
        }
    }
    self.clone()
}
```

1.2 AscentFlip

Reverse of DescentFlip. Walk strata $l = 0, 1, \dots, \text{level} - 1$ and $k = 1, \dots, \text{level}$. Pick the largest atom a such that ka is an Apéry element of S landing in stratum l and $ka \neq f + m$. Remove a from the generators (equivalent to toggling a).

```
pub fn ascent_flip(&self) -> Self {
    // Apéry membership is  $O(1)$  via residue indexing:
    //  $v \in \text{apery\_set} \iff \text{apery\_set}[v \bmod m] == v$ .
    let is_apery = |v: usize| self.apery_set[v % self.m] == v;
    for l in 0..self.level {
        for k in 1..self.level {
            let removable = |a: usize| -> bool {
                let v = k * a;
                v > self.m && v + l * self.m < self.f && v + (l + 1) * self.m > self.f;
            };

```

```

        if let Some(w) = self
            .gen_set
            .iter()
            .copied()
            .filter(|&a| removable(a) && is_apery(k * a))
            .max()
        {
            return self.toggle(w);
        }
    }
    self.clone()
}

```

1.3 DescentTotal

Walk strata $l = 1, 2, \dots$ and stop at the first l at which the open interval $(f - (l+1)m, f - lm)$ is *not* fully in S . Candidate new generator: $x := f - (l-1)m$ (a gap of residue μ). Commit the candidate only if applying AscentTotal to the result returns S ; otherwise no-op. The safety gate rejects two situations: (i) μ shifts because some w_i with $i \neq \mu$ exceeds x ; (ii) a residue- μ generator $g > x$ of S becomes reducible as $g = x + km$ and is dropped from the result.

```

pub fn descent_total(&self) -> Self {
    if self.f < self.m {
        return self.clone();
    }
    // Stratum- $l$  predicate: returns true iff  $(f - (l+1)m, f - lm) \subset S$ .
    // For  $l$  in  $1..=level$  we have  $l \cdot m \leq level \cdot m < f$ , so the upper bound
    //  $f - l \cdot m$  is always strictly positive; the lower bound can underflow
    // when  $(l+1) \cdot m > f$ , in which case we clamp to 0 (the interval still
    // makes sense as  $(0, f - l \cdot m)$ ).
    let stratum_in_s = |l: usize| -> bool {
        let upper = self.f - l * self.m;
        let lower = (l + 1)
            .checked_mul(self.m)
            .and_then(|p| self.f.checked_sub(p))
            .unwrap_or(0);
        ((lower + 1)..upper).all(|y| self.element(y))
    };
    for l in 1..=self.level {
        if !stratum_in_s(l) {
            let candidate = self.toggle(self.f - (l - 1) * self.m);
            // Commit only when ascent_total cleanly inverts; otherwise
            // the would-be descent is a -shift or a gen-collapse and
            // we leave self unchanged.

```

```

        if candidate.ascent_total() == *self {
            return candidate;
        }
        return self.clone();
    }
}
self.clone()
}

```

1.4 AscentTotal

Destroy the atom $f + m$ when it lies in the generator set, i.e. return $\langle \text{Atom}(S) \setminus \{f + m\} \rangle$.
No-op when $f + m \notin \text{Atom}(S)$.

```

pub fn ascent_total(&self) -> Self {
    let ae = self.f + self.m;
    if self.gen_set.contains(&ae) {
        self.destroy_atom(ae)
    } else {
        self.clone()
    }
}

```

1.5 Effect on (g, σ, r) and on e

	DescentFlip	DescentTotal	AscentFlip	AscentTotal
g'	$g - 1$	$g - l$	$g + 1$	$g + \rho$
σ'	$\sigma + 1$	$\sigma + l$	$\sigma - 1$	$\sigma - \rho$
r'	$r - 2$	$r - 2l$	$r + 2$	$r + \rho(m - 2)$
e' if $f + m \in \text{Atom}(S)$	?	?	?	?
e' if $f + m \notin \text{Atom}(S)$	?	?	?	?

The (g, σ, r) entries follow from $r + \sigma = g$ and the fact that DescentFlip and AscentFlip each flip a single reflected-gap pair. The total operations bundle l (resp. ρ) flips into a single step.

Open question on $e' = e \pm \delta$ when $f + m$ is or is not an atom: e.g. DescentTotal can pull a residue- μ atom down to the new $f + m$ position, making the old $f + m$ reducible ($\sim = x + km$), so δ can be negative; the count depends on how many residue- μ atoms lay above the new x . AscentTotal removes one atom by construction but may make several previously- expressible elements newly minimal. The exact counts are left as follow-up work and tagged "?" in the table.

2 Setup

Let S be a numerical semigroup, with f, m, g, σ as usual (Frobenius, multiplicity, genus, number of sporadic elements). Write $S_+ := S \setminus \{0\}$.

- $\text{Ap}(S) := \{s \in S \mid s - m \notin S\}$, the *Apéry set* (one representative per residue class mod m). Index it by residue: $\text{Ap}(S) = \{w_0, \dots, w_{m-1}\}$ with $w_i \equiv i \pmod{m}$. In particular $w_0 = 0$.
- $\text{RG}(S) := \{L \in \{1, \dots, f-1\} \mid L \notin S, f-L \notin S\}$, the *reflected gaps*.
- a is an *atom* of S iff $a \in S_+ \setminus (S_+ + S_+)$. Equivalently, a lies in the unique minimal generating set $\text{Atom}(S)$ of S .
- $\mu \in \{1, \dots, m-1\}$ denotes $\mu := f \bmod m$; equivalently $f + m = w_\mu$ (the largest Apéry element).
- For each $i \in \{1, \dots, m-1\}$, set

$$r_i := \frac{w_i + w_{\mu-i} - w_\mu}{m},$$

where the index $\mu-i$ is taken mod m . Geometrically: $w_i + w_{\mu-i}$ sits exactly r_i steps of m above w_μ on the residue- μ ladder, so the closed interval $[w_\mu, w_i + w_{\mu-i}]$ in that ladder contains $r_i + 1$ rungs (the rung $w_\mu - m = f$ being a gap and the rest of the rungs above w_μ being elements of S). r_i counts the reflected gaps in residue class $i \pmod{m}$.

- $V(S, l) := \{f - lm + 1, f - (l-1)m + 2, \dots, f-1\}$, the l -Stratum, for $l = 1$ this is the *interval below f* (always $m-1$ consecutive integers).
- $\rho(S) := \min_{i \in \{1, \dots, m-1\} \setminus \{\mu\}} r_i$. The class $i = \mu$ is excluded because $r_\mu = 0$ trivially (every $x \equiv \mu$ has $f - x \equiv 0 \pmod{m}$, hence $f - x \in S$).
- Total reflected-gap count: $r = \sum_{i=1}^{m-1} r_i = \#\text{RG}(S)$ (the term r_0 would equal 0 anyway).
- Trivial identities: $r + \sigma = g$, $f + 1 + r = 2g$, and $t - 1 \leq r$. The first follows from closure: no pair $\{x, f-x\}$ can lie inside S_+ (else $f \in S$), and similarly $f/2$ is always a gap when f is even; pairing up $\{0, 1, \dots, f\}$ under $x \leftrightarrow f-x$ then gives $r + \sigma = g$. The second is $g + \sigma = f + 1$ combined with the first. The third holds because each pseudo-Frobenius $z \neq f$ is a reflected gap ($z + S_+ \subseteq S$ would imply $f \in S$ if $f - z \in S_+$).

3 UpDown

Fix $m > 2$. We define four families of one-step moves, two *descents* ($S \mapsto S^\downarrow$, adding one element) and two *ascents* ($S \mapsto S^\uparrow$, removing one atom). Primes denote the new counts in $S^\downarrow$ resp. $S^\uparrow$. The descents are exactly the inverses of the ascents ($\mathcal{A}_{flip}$ inverts $\mathcal{D}_{flip}$, with $w_i \in V(S)$ playing the role of $w_i - m$; $\mathcal{A}_{total}$ inverts $\mathcal{D}_{total}$).

1. $\mathcal{D}_{flip}(i)$, with $i \in \{1, \dots, m-1\} \setminus \{\mu\}$ and $f < w_i < f + m$:

$$S^\downarrow := S \cup \{w_i - m\}, \quad g' = g - 1, \quad \sigma' = \sigma + 1, \quad r' = r - 2.$$

2. $\mathcal{D}_{total}(\mu)$, requiring $V(S) \subseteq S$:

$$S^\downarrow := S \cup \{w_\mu - m\} = S \cup \{f\}, \quad g' = g - 1, \quad \sigma' = \sigma - (m-1), \quad r' = r + (m-2).$$

3. $\mathcal{A}_{flip}(i)$, with $i \in \{1, \dots, m-1\} \setminus \{\mu\}$, $ka = w_i$ and Apéry-Element for an atom a of S and $k > 0$, and $w_i \in V(S, l)$ (going down on l if necessary):

$$S^\uparrow := S \setminus \{a\}, \quad g' = g + 1, \quad \sigma' = \sigma - 1, \quad r' = r + 2.$$

4. $\mathcal{A}_{total}(\mu)$, requiring $w_\mu = f + m$ to be an atom of S (equivalently $\rho(S) > 0$); set $\rho := \rho(S)$:

$$S^\uparrow := \langle A(S) \setminus \{w_\mu\} \rangle, \quad g' = g + \rho, \quad \sigma' = \sigma + \rho(m-1), \quad r' = r - \rho(m-2).$$

Note: $S^\uparrow$ is *not* the set $S \setminus \{w_\mu\}$. The latter happens to be closed under $+$ already (when $\rho > 0$) but only drops one element; the structural ascent removes $w_\mu, w_\mu + m, \dots, w_\mu + (\rho-1)m$ in one shot, since those rungs of the residue- μ ladder were reachable in S only via w_μ . The first rung of class μ that survives is $w_\mu + \rho m = w_i + w_{\mu-i}$ for some $i \in \{1, \dots, m-1\} \setminus \{\mu\}$.

All four claims are proved by tracking the Apéry set under the move. Aside: the Wilf conjecture in the form $(e-2)\sigma \geq r$ (which is equivalent to the standard $e\sigma \geq f+1$ via $r+\sigma=g$ and $g+\sigma=f+1$) constrains σ and r in tandem. We know the change of both in every case, so working out the change of e would close the loop.

4 Correspondence

We now zoom in on the total cases. Fix $m > 2$ throughout this section.

Equivalence: w_μ **atom** $\Leftrightarrow \rho(S) > 0$. If $\rho > 0$ then $w_i + w_{\mu-i} = w_\mu + r_i m > w_\mu$ for every $i \in \{1, \dots, m-1\} \setminus \{\mu\}$, so w_μ is not a sum of two positive Apéry elements, hence (since every element of S_+ is $w_j + km$ for some j and $k \geq 0$) not a sum of two elements of S_+ at all; that is, w_μ is an atom. Conversely, if some $r_i = 0$ then $w_i + w_{\mu-i} = w_\mu$ with both summands in S_+ , so w_μ is decomposable.

When can $\mathcal{D}_{total}$ and $\mathcal{A}_{total}$ undo each other? Suppose we start at S with $\rho(S) = 0$, so $w_\mu(S)$ is not an atom and we cannot apply $\mathcal{A}_{total}$ to S directly. Apply $\mathcal{D}_{total}$ instead (assuming $V(S) \subseteq S$): we obtain $S^\downarrow := S \cup \{f\}$, whose Frobenius drops and whose new w_μ is the atom f (a fresh minimal generator). If the new $\rho(S^\downarrow) = 1$, then applying $\mathcal{A}_{total}$ to $S^\downarrow$ removes exactly one element — the new w_μ , namely the original f — and returns us to S . This is the one situation in which a single descent followed by a single ascent is a round trip:

$$S_{\rho=0} \xrightarrow{\mathcal{D}_{total}} D(S)_{\rho=1} \xrightarrow{\mathcal{A}_{total}} S.$$

For $\rho(D(S)) > 1$ the round trip is no longer one-step in the ascent: $\mathcal{A}_{total}$ deletes ρ elements at once, so to return to S we would need ρ separate descents (each peeling off one element). Concretely, undoing the ascent's $(g + \rho, \sigma + \rho(m-1), r - \rho(m-2))$ by ρ applications of $\mathcal{D}_{total}$ gives a net change of

$$g + \rho - \rho \cdot 1 = g, \quad \sigma + \rho(m-1) - \rho(m-1) = \sigma, \quad r - \rho(m-2) + \rho(m-2) = r,$$

as required. But each individual descent step demands $V \subseteq S$ at that point, which may fail before the round trip completes; see the next subsection.

The map D . Let $\mathbb{S}_\mu^{\text{old}}$ be the set of numerical semigroups S with $m(S) = m$ and $\mu(S) = \mu$, and satisfying $f(S) > m$ and $V(S) \subseteq S$. Define

$$D : \mathbb{S}_\mu^{\text{old}} \longrightarrow \{\text{numerical semigroups with } m(S) = m\}, \quad D(S) := S \cup \{f(S)\}.$$

(We do *not* claim $D(S)$ has Frobenius congruent to $\mu \bmod m$; in general $\mu(D(S)) \neq \mu(S)$. The subscript “ μ ” refers to the residue of f in S , not in $D(S)$.) The following properties are useful and are proved by tracking the Apéry set through the descent:

- $\rho(D(S)) = \rho(S) + 1$, where on the left ρ is computed with respect to $\mu(D(S))$.
- In residue class i relative to the new $\mu(D(S))$, $r_i(D(S)) = r_i(S) + 1$.
- $\rho(D(S)) > 0$, so $w_\mu(D(S)) = f(S) + m$ is an atom of $D(S)$, i.e. $D(S)$ is eligible for $\mathcal{A}_{\text{total}}$.

What remains to control along an iterated descent is whether $V(D(S)) \subseteq D(S)$ — the condition needed to apply $\mathcal{D}_{\text{total}}$ a second time.

4.1 One-step bijection

The only way a single $(\mathcal{D}_{\text{total}}, \mathcal{A}_{\text{total}})$ pair is a round trip is

$$S_{\rho=0} \longleftrightarrow D(S)_{\rho=1},$$

i.e. when $w_\mu(S)$ is *not* an atom of S but $D(S)$ has $\rho = 1$. Only then is $D(S) \in \mathcal{A}_{\text{total}}(\mu(D(S)))$ and the ascent’s removal-of- ρ -elements coincides with the descent’s addition of f .

Symmetric semigroups ($r = 0$) are excluded from $\mathcal{A}_{\text{total}}$: $r = 0$ forces every $r_i = 0$, hence $\rho = 0$, hence w_μ is not an atom.

4.2 $f + m$ atom: iterated descent, $\rho(S) > 0$

As long as $V(S) \subseteq S$ we can keep descending:

$$S_\rho \rightarrow D(S)_{\rho+1} \rightarrow D^2(S)_{\rho+2} \rightarrow \dots$$

At every step $f + m$ is the new atom w_μ (so $\mathcal{A}_{\text{total}}$ remains applicable), and ρ increases by 1. The chain terminates exactly when either $V(S) \not\subseteq S$ at the current S (so $\mathcal{D}_{\text{total}}$ no longer applies) or $f < m$ (so $V(S)$ is empty/degenerate).

5 Code

```
fn test_up_downs(s: &Semigroup) {
    if s.m == 1 {
        return;
    }
    for i in 1..s.m {
        let w = s.apery_set[i];
```

```

// Descent: w-m is a gap; adding it lifts the membership of w-m.
if w > s.f && s.f > s.m {
  assert!(!s.element(w - s.m));
  assert!(w <= s.f + s.m);
  assert!(
    w == s.f + s.m || s.is_reflected_gap(w - s.m),
    "reflected {w} at {i}"
  );
  let down = s.toggle(w - s.m);
  assert_eq!(down.g + 1, s.g);
  if w == s.f + s.m && s.v_in_s() {
    assert_eq!(down.r + 2, s.r + s.m, "r change w==f+m for i={i}");
    assert_eq!(down.sigma + s.m, s.sigma + 1, "sigma");
  }
  if w < s.f + s.m {
    assert_eq!(down.r + 2, s.r, "r change w<f+m for i={i}");
    assert_eq!(down.sigma, s.sigma + 1, "sigma");
  }
}

// Ascent: w must already be a minimal generator.
if s.gen_set.contains(&w) {
  let up = s.toggle(w);
  if w < s.f && w + s.m > s.f {
    assert_eq!(
      up.g,
      s.g + 1,
      "removing atom generates gap for {i} and {w}"
    );
    assert!(up.apery_set.contains(&(w + s.m)));
    assert_eq!(up.apery_set[i], w + s.m);
    assert_eq!(up.r, s.r + 2, "r change w<f for i={i}");
    assert_eq!(up.sigma + 1, s.sigma);
  }
  if w == s.f + s.m {
    let rho = s.rho();
    assert_eq!(
      up.g,
      s.g + rho,
      "removing atom generates rho gap for {i} and {w}"
    );
    assert!(up.apery_set.contains(&(w + rho * s.m)));
    assert_eq!(up.apery_set[i], w + rho * s.m);
    assert_eq!(up.r + rho * (s.m - 2), s.r, "r change w==f+m for i={i}");
    assert_eq!(

```

```

        up.sigma,
        rho * (s.m - 1) + s.sigma,
        "sigma change w==f+m for i={i}"
    );
}
}
}
}
}

```

6 Claude's review notes

What this writeup is about, in plain words

You have a numerical semigroup S with multiplicity m and Frobenius number f . Its Apéry set has one element w_i per residue class mod m , and the largest of these is $w_\mu = f + m$ where $\mu = f \bmod m$. The writeup studies four local *moves* that turn S into a nearby semigroup:

$\mathcal{D}_{flip}$ (**descent, flip**): Pick an Apéry element w_i that lies in the open interval $(f, f + m)$ with $i \neq \mu$. Add $w_i - m$ to S . This fills in a reflected gap and lowers w_i by m .

$\mathcal{D}_{total}$ (**descent, total**): When the whole interval $V(S) = \{f - m + 1, \dots, f - 1\}$ already lies in S , add f itself. This collapses the Apéry structure dramatically: the new Frobenius drops below $f - m$, many Apéry elements get re-indexed at once, and σ decreases by $m - 1$.

$\mathcal{A}_{flip}$ (**ascent, flip**): The exact inverse of $\mathcal{D}_{flip}$: remove a generator $w_i \in V(S)$ with $i \neq \mu$.

$\mathcal{A}_{total}$ (**ascent, total**): The harder one. Remove $w_\mu = f + m$ as a generator. Because w_μ is the only way to reach the next few rungs of its residue ladder, you actually lose ρ elements in one move, where ρ is the smallest r_i over $i \neq \mu$.

The book-keeping (how g , σ , r change in each case) is mostly mechanical. What is interesting is whether you can *compose* these moves in a controlled way: take a descent followed by an ascent and ask when you get back to the same semigroup. That is the content of the “Correspondence” section. The punchline is: a single descent–ascent pair is a round trip only when the descent target has $\rho = 1$, i.e. the descent lifts a semigroup with w_μ *not* an atom to one with w_μ a uniquely-decodable atom. For higher ρ , you can iterate descents (provided $V(S) \subseteq S$ keeps holding), with each step increasing ρ by one and keeping $f + m$ an atom — until the chain hits a wall, namely a gap landing inside $V(S)$ or f dropping below m .

In short, this is a structural “hill-climbing” picture of the descent/ascent dynamics on the Kunz/Apéry side of the Wilf conjecture, with the four moves serving as elementary edges of an operation graph on numerical semigroups.

Status of the issues from the first pass

- **Case 4 definition.** Fixed: now $\langle A(S) \setminus \{w_\mu\} \rangle$, with a paragraph clarifying why the naive set-minus is the wrong object.
- $\rho > 0$ **hypothesis in Case 4.** Fixed: stated inline as “equivalently $\rho(S) > 0$ ”, and the equivalence is proved in the Correspondence section.
- **Index inconsistency** $\mathcal{D}_{total}(i, \mu)$ **vs.** $\mathcal{A}_{total}(i, m)$. Fixed: the spurious second argument is gone everywhere; both families take only μ .
- **Codomain of D .** Fixed: D now lands in “numerical semigroups with multiplicity m ” rather than $\mathbb{S}_\mu$. The old $\mathbb{S}_\mu$ has been renamed $\mathbb{S}_\mu^{\text{old}}$ and serves only as the domain of D .
- $r + \sigma = g$ **justification.** Fixed: a one-sentence reason (no $\{x, f - x\} \subset S_+$ by closure, pair-counting on $[0, f]$) appears in the Setup section.
- **“Viewed from below” equations.** Fixed: rewritten as ρ descents undoing one ascent, with the matching arithmetic.
- **Minor wording / typos.** Fixed: “necessesay / beeing / aways / The the / Semi-group”, stray standalone S , range of L in RG, two Apéry definitions, ambiguous sum range, tautological “minimal generator” parenthetical, Wilf parenthetical.

Remaining issues (still open)

1. **No formal proofs.** The four change-formulas in Section UpDown are stated and informally justified (“proved by tracking the Apéry set”). The codebase property test `test_up_downs` checks them empirically over a wide fixture set, so they are very likely correct, but a reader of the writeup gets no algebraic derivation. Worth writing out at least one case (the $\sigma' = \sigma + \rho(m - 1)$ case is the only one that surprises) as a small lemma.
2. $\rho(D(S)) = \rho(S) + 1$ **asserted, not proved.** The Correspondence section still says “the following properties are useful” without supplying arguments. The `s.r_i(i)` identity for Kunz coefficients is the natural proof tool, but the writeup doesn’t reference it.
3. $V(D(S)) \subseteq D(S)$ **termination condition.** The iterated descent chain is described informally as “stops once there is a gap in V or $f < m$ ”. There is presumably a clean characterisation of which semigroups admit a chain of length $\geq k$, but the writeup doesn’t attempt one. Could be stated either combinatorially (in terms of the Apéry set) or via the Kunz matrix.
4. **Symmetry / direction conventions.** “Descent” *adds* elements to S ; “ascent” *removes* them. This convention (descend = grow) is consistent with the codebase but is the reverse of what some readers might expect from the words. A one-line note up front (“descent shrinks the gap set; ascent enlarges it”) would help.

5. **Wilf-aside is a teaser, not a result.** The line “we really should work out the change in e ” is a TODO; no statement about how e moves under any of the four operations is given. If the writeup is meant to be self-contained even as an informal note, this is the most natural next section.
6. **Code listing is not commented on.** The verbatim Rust test in Section Code is given without text explaining which assertion corresponds to which case in Section UpDown. Adding case labels in comments (`// Case 1: D_flip`, etc.) would help the reader who wants to cross-check the math against the code.
7. **Where $\mathcal{D}_{flip}$ / $\mathcal{A}_{flip}$ live in the descent graph.** Section Correspondence treats only the *total* pair. Whether the flip pair admits any analogous round-trip discussion is not addressed. The inversion remark “ $\mathcal{A}_{flip}$ inverts $\mathcal{D}_{flip}$ ” is a single sentence; the consequence (that the flip pair is always a round-trip on the appropriate domain) deserves a paragraph.
8. **Notation drift between r_i and $c(i, j)$.** Internally to the codebase, $r_i = c(i, \mu - i)$ where $c(\cdot, \cdot)$ is the Kunz coefficient. The writeup never introduces c , even though the Kunz interpretation makes some arguments (especially $\rho(D(S)) = \rho(S) + 1$) crisper.

Notation $\leftrightarrow$ codebase

LaTeX	Code
m, f, μ	<code>s.m, s.f, s.mu</code>
g, σ, r	<code>s.g, s.sigma, s.r</code>
r_i, ρ	<code>s.r_i(i), s.rho()</code>
$w_i, \text{Ap}(S)$	<code>s.apery_set[i], s.apery_set</code>
$\text{RG}(S)$	<code>s.blob()</code>
$V(S) \subseteq S$	<code>s.v_in_s()</code>
atom	element of <code>s.gen_set</code>
$S^\downarrow$ (descent)	<code>s.toggle(w - s.m)</code>
$S^\uparrow$ (ascent)	<code>s.toggle(w)</code>
t	<code>s.t</code> (Rust) / <code>s.type_t</code> (JS)